NVIDIA RTX DLSS Neural Rendering UE Plugin

Table of Contents

- Overview
- Requirements and Supported Engine Versions
- Main Features and Controls
  - Global Controls
  - Local Controls Using Stencil Mask
  - Local Control Using Auto Mask
- Workflow
  - Local Stencil Mask
    - 1. Enable Custom Depth/Stencil
    - 2. Create the Local Mask Library
    - 3. Enable DLSS-NR and Stencil Mask Mode
    - 4. Assign Stencil IDs to Primitives
    - 5. Inspect Assigned IDs with Buffer Visualization
    - 6. Inspect the Generated DLSS-NR Local Mask
- Technical Reference: CVars
- Technical Reference: Blueprint Nodes

Overview

NVIDIA RTX DLSS Neural Rendering (DLSS-NR) is a Streamline feature that applies a photoreal image uplift to the rendered frame. In this Unreal Engine plugin, DLSS-NR is exposed through runtime console variables, Blueprint nodes, and Streamline project settings for optional local masking.

Requirements and Supported Engine Versions

- Unreal Engine: the plugin is developed and tested with UE version 5.5 to 5.8 inclusive.
- Platform: Win64 (x86_64).
- RHI: DirectX 12.
- GPU: Blackwell or later.
- NVIDIA Driver: 615.85

Main Features and Controls

Global Controls

Global controls apply to the full DLSS-NR output unless a local mask attenuates the effect.

- **Style** selects one of the discrete DLSS-NR looks. Supported style IDs are `0`, `1`, and `2`; evaluate styles first before tuning strength values.
- **Tone Strength** controls the degree of tonal/luminance remapping toward the NR reference look.
- **Structural Strength** controls enhancement of fine structure such as surface detail, edges, shadows, and microstructure.
- **Intensity** controls the overall blend between the original input and the NR-uplifted output. For best results it's recommended to tune tone strength and structural strength first before resorting to intensity. `0` is pass-through; `1` is full effect.

Local Controls Using Stencil Mask

Stencil mask mode uses each primitive's CustomDepth Stencil Value to look up a mask setting structure in the Streamline `Stencil Mask Settings`. The setting structure is converted into a four-channel DLSS-NR mask. The primary content-authoring controls are:

- **Local Tone Strength**: local multiplier for tone strength.
- **Local Structural Strength**: local multiplier for structural strength.
- **Intensity**: local multiplier for global intensity.

The settings object also exposes **Global Tone Strength** in the local mask settings which is deprecated in the current streamline SDK.

Objects with unconfigured stencil values receive the default setting `1.0, 1.0, 1.0, 1.0`. The default mask setting can be changed with CVars `r.Streamline.DLSSNR.Mask.DefaultStencil*`.

Local Control Using Auto Mask

The auto mask mode currently target is skin. Use this when the project needs a broad skin-specific structural adjustment without authoring stencil IDs. Auto mask is selected with `r.Streamline.DLSSNR.Mask.Mode=2` or the `Set DLSS-NR Control Mask Mode` Blueprint node set to `Auto Mask`. The **Skin Structure Strength**. is controlled with `r.Streamline.DLSSNR.Mask.SkinStructureStrength` cvar.

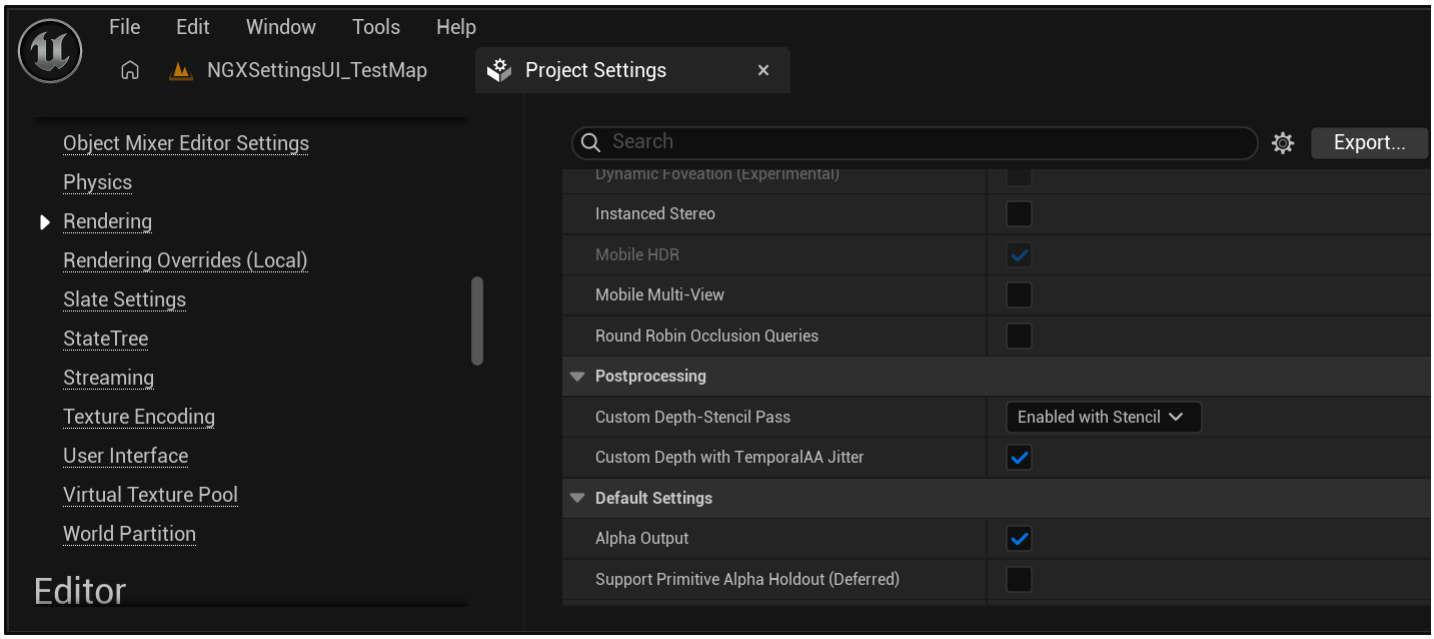
Workflow

1. Pick the style.
2. Tune structural and tone strength.
3. Tune local stencil or auto-mask controls only where global controls are not enough.

Local Stencil Mask

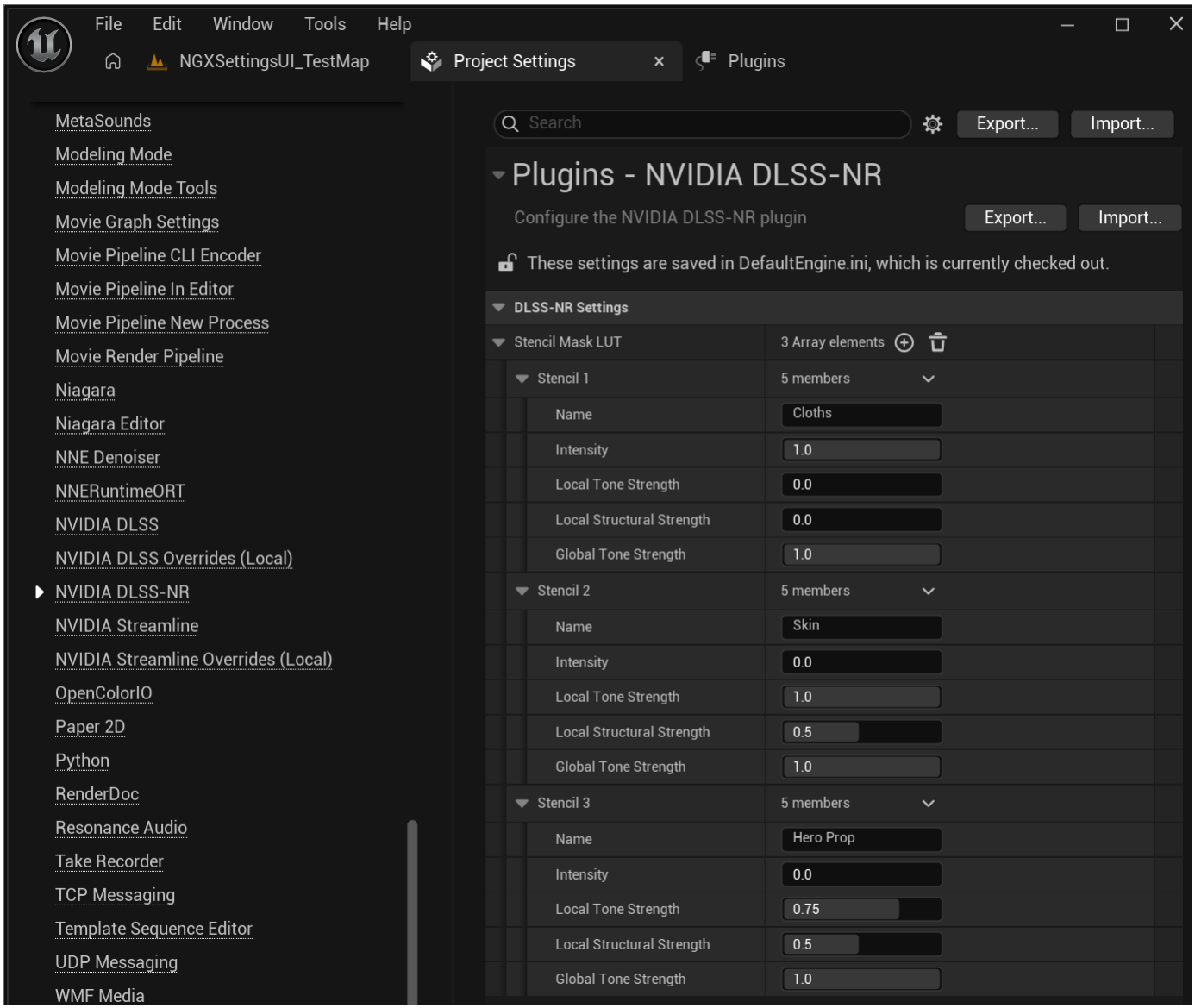
1. Enable Custom Depth/Stencil

Open `Project Settings -> Engine -> Rendering` and set **Custom Depth-Stencil Pass** to **Enabled with Stencil**. This writes `r.CustomDepth=3` in project config.



2. Create the Local Mask Library

Open `Project Settings -> Plugins -> NVIDIA DLSS-NR` and edit **DLSS-NR Settings -> Stencil Mask Settings**. Each array entry maps directly to a stencil ID by index.



Example mapping:

Stencil ID	Example name	Intended primitive set
1	Skin	Characters, faces, hands
2	Cloth	Cloth, fabric, outfits
3	Hero Prop	Important props that need separate tuning

3. Enable DLSS-NR and Stencil Mask Mode

Use Blueprint or cvars:

```
r.Streamline.DLSSNR.Enable=1
r.Streamline.DLSSNR.Mask.Mode=1
```

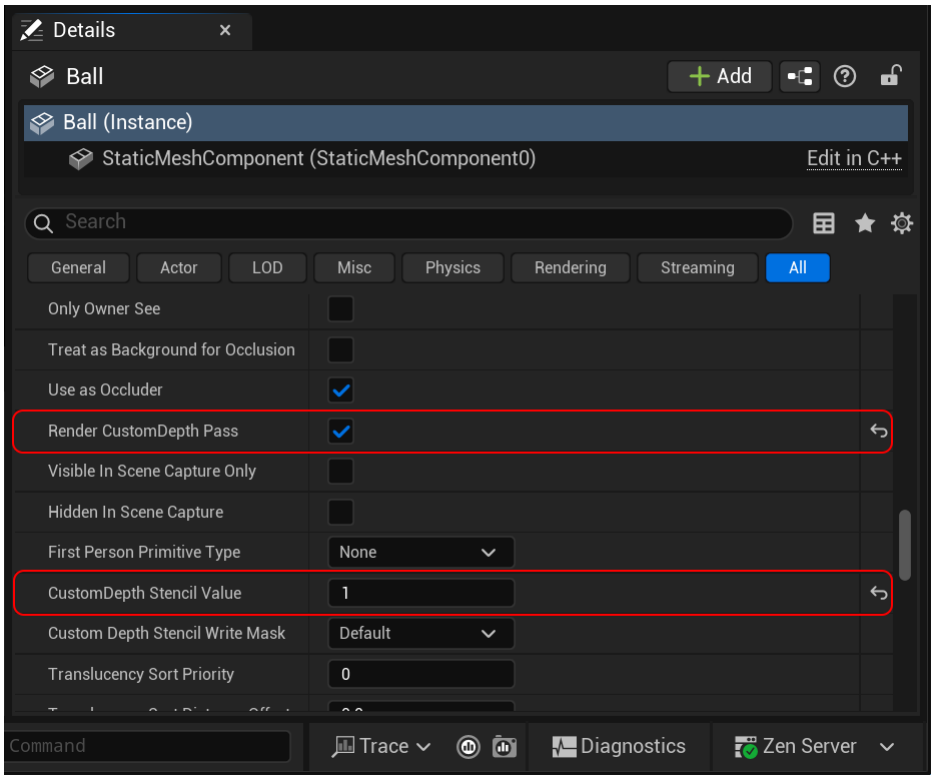
`Mask.Mode=1` selects Local Stencil Mask. The generated mask is only produced when this mode is active.

4. Assign Stencil IDs to Primitives

Select a primitive in the level. In the Details panel under **Rendering**:

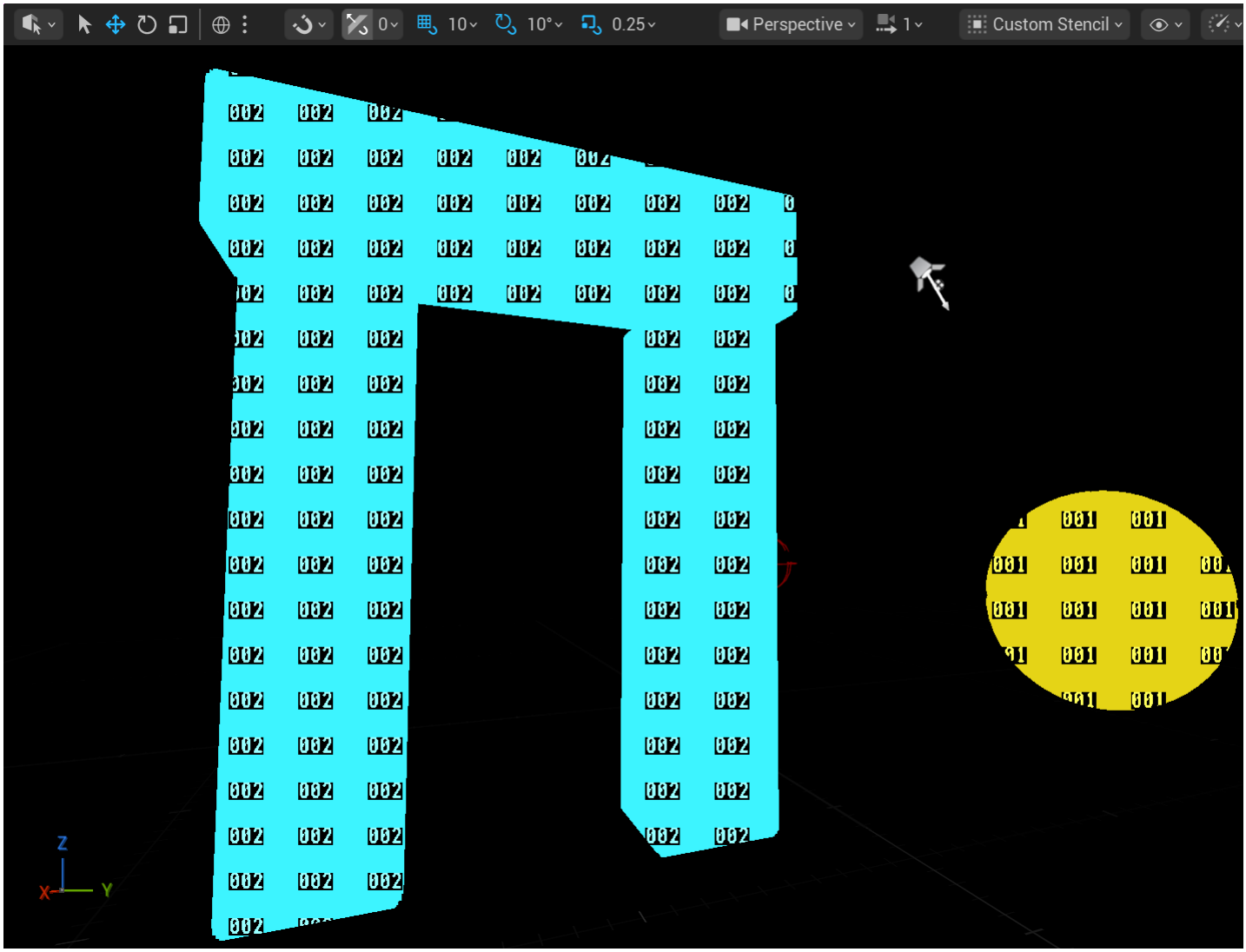
1. Enable **Render CustomDepth Pass**.
2. Set **CustomDepth Stencil Value** to the stencil ID you want.
3. Leave **CustomDepth Stencil Write Mask** at the default full mask unless your project has a special reason to change it.

For example, setting `CustomDepth Stencil Value` to `1` maps the primitive to stencil entry `ID 1`.



5. Inspect Assigned IDs with Buffer Visualization

Use the editor viewport menu `Buffer Visualization -> Custom Stencil` to visually identify the stencil IDs assigned to each primitive. This is useful for confirming that each primitive maps to the intended stencil row.

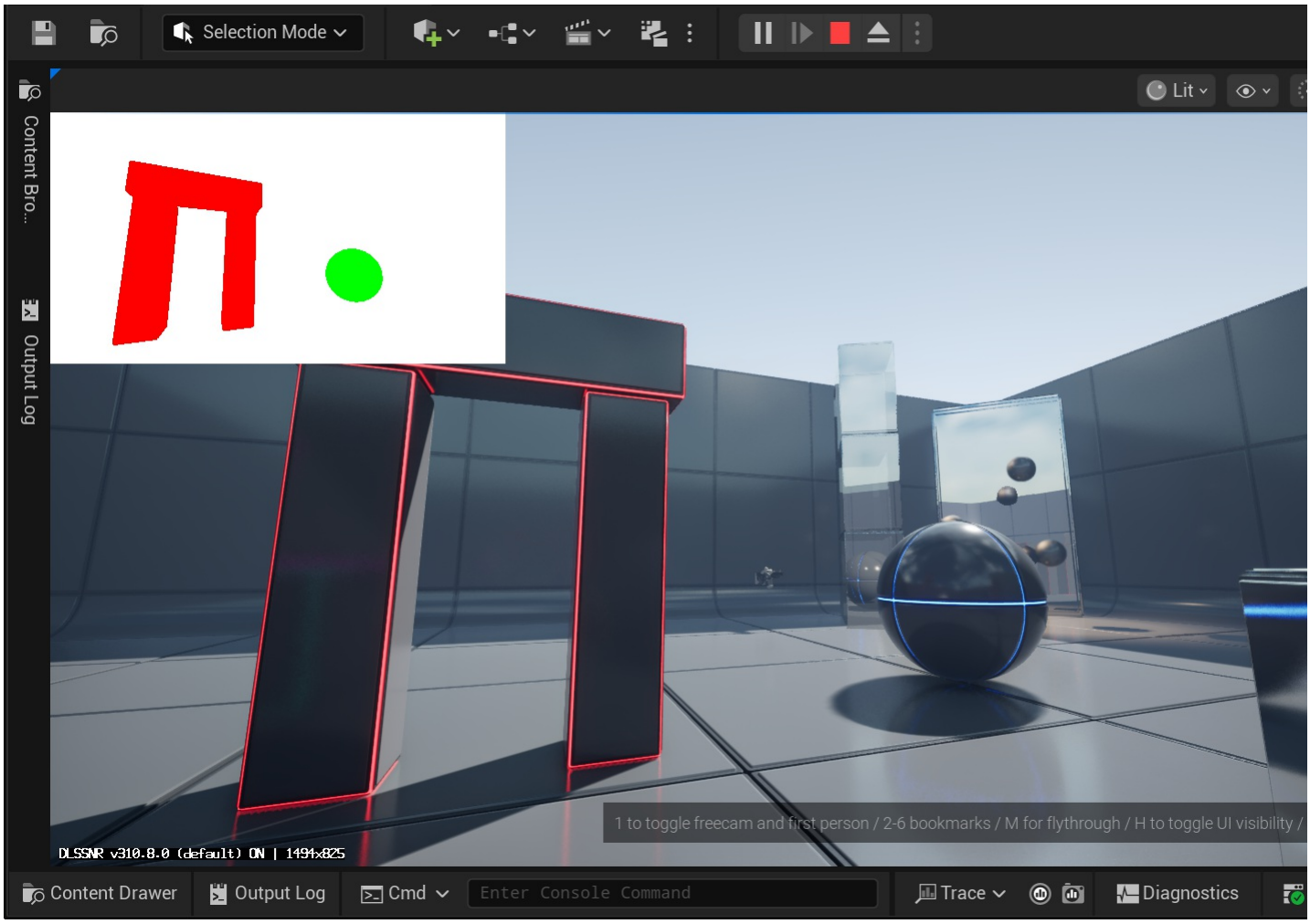


6. Inspect the Generated DLSS-NR Local Mask

Enable the local mask visualization overlay:

```
r.Streamline.DLSSNR.Mask.VisualizeStencil=1
```

This requires DLSS-NR to be active and `r.Streamline.DLSSNR.Mask.Mode=1`. The plugin draws the generated mask as a scaled overlay in the top-left of the view, so you can verify the local mask values applied to each primitive.



Disable it when finished:

```
r.Streamline.DLSSNR.Mask.VisualizeStencil=0
```

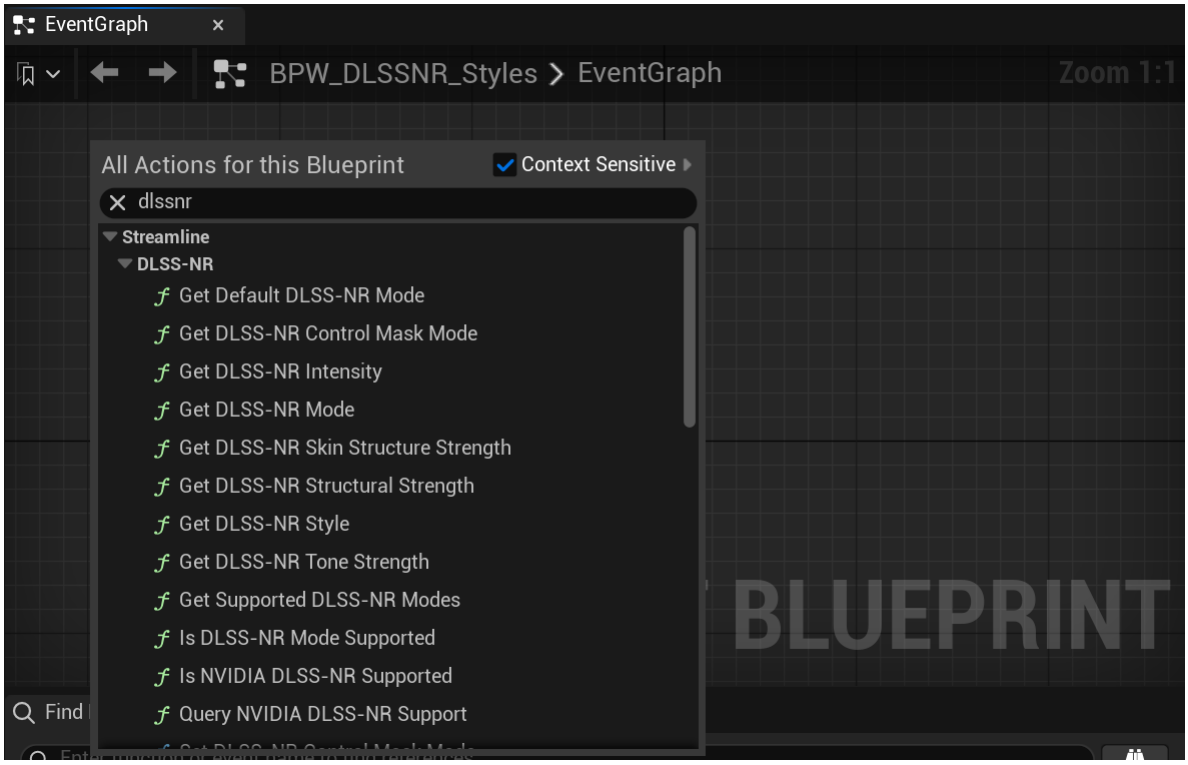
Technical Reference: CVars

CVar	Default	Description
`r.Streamline.DLSSNR.Enable`	`0`	Enables DLSS-NR. `0` = off, `1` = on.
`r.Streamline.DLSSNR.Style`	`0`	Selects DLSS-NR style. Supported IDs are `0`, `1`, and `2`.
`r.Streamline.DLSSNR.ToneStrength`	`1.0`	Controls tonal/luminance remapping strength.
`r.Streamline.DLSSNR.StructuralStrength`	`1.0`	Controls structure/detail enhancement strength.
`r.Streamline.DLSSNR.Intensity`	`1.0`	Overall blend strength. `0` keeps the input unchanged; `1` applies the full NR effect.
`r.Streamline.DLSSNR.Mask.Mode`	`0`	Selects masking. `0` = off, `1` = local stencil mask, `2` = auto mask.
`r.Streamline.DLSSNR.Mask.SkinStructureStrength`	`1.0`	Structural strength for skin regions when auto mask is active.
`r.Streamline.DLSSNR.Mask.DefaultStencilIntensity`	`1.0`	Intensity component used for objects with unconfigured stencil values.

CVar	Default	Description
`r.Streamline.DLSSNR.Mask.DefaultStencilToneStrength`	`1.0`	Local tone strength component used for objects with unconfigured stencil values..
`r.Streamline.DLSSNR.Mask.DefaultStencilStructuralStrength`	`1.0`	Local structural strength component used for objects with unconfigured stencil values.
`r.Streamline.DLSSNR.Mask.VisualizeStencil`	`0`	Draws the generated stencil mask overlay in the top-left of the view. Requires mask mode `1`.
`r.Streamline.DLSSNR.Mask.OcclusionDepthTolerance`	`0.01`	Relative depth tolerance used when rejecting stencil-mask pixels hidden behind closer scene geometry.

Technical Reference: Blueprint Nodes

Blueprint category: `Streamline | DLSS-NR`.



Node	Description
`Is NVIDIA DLSS-NR Supported`	Returns true only when DLSS-NR is supported on the current platform, RHI, GPU, and driver.
`Query NVIDIA DLSS-NR Support`	Returns the detailed Streamline feature support enum. Use this to distinguish unsupported platform, RHI, or hardware cases.
`Is DLSS-NR Mode Supported`	Checks whether `Off` or `On` is valid in the current runtime state.
`Get Supported DLSS-NR Modes`	Returns the supported `EStreamlineDLSSNRMode` values for UI population.
`Set DLSS-NR Mode`	Sets `r.Streamline.DLSSNR.Enable`. `Off` disables DLSS-NR; `On` enables it.
`Get DLSS-NR Mode`	Reads the current mode from `r.Streamline.DLSSNR.Enable`.
`Get Default DLSS-NR Mode`	Returns the plugin default mode. In this implementation the default is `Off`.
`Set DLSS-NR Style` / `Get DLSS-NR Style`	Writes or reads `r.Streamline.DLSSNR.Style`.
`Set DLSS-NR Tone Strength` / `Get DLSS-NR Tone Strength`	Writes or reads `r.Streamline.DLSSNR.ToneStrength`.
`Set DLSS-NR Structural Strength` / `Get DLSS-NR Structural Strength`	Writes or reads `r.Streamline.DLSSNR.StructuralStrength`.
`Set DLSS-NR Intensity` / `Get DLSS-NR Intensity`	Writes or reads `r.Streamline.DLSSNR.Intensity`.
`Set DLSS-NR Control Mask Mode` / `Get DLSS-NR Control Mask Mode`	Writes or reads `r.Streamline.DLSSNR.Mask.Mode`. Values are `Off`, `Stencil Mask`, and `Auto Mask`.

Node	Description
`Set DLSS-NR Skin Structure Strength` / `Get DLSS-NR Skin Structure Strength`	Writes or reads `r.Streamline.DLSSNR.Mask.SkinStructureStrength`. Use with auto mask.